

Kunli Liu, PhD

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SUMMARY

Software/infrastructure leader specializing in large-scale distributed systems for reliability, observability, and scalability. Owns end-to-end platform architecture with measurable impact on developer velocity and revenue loss prevention. Experienced leading cross-org engineering initiatives and driving technical direction from ideation to production rollout.

Languages: Hack/PHP, C++, Python

Engineering Leadership: multi-team roadmap planning, cross-org partner alignment, headcount & capacity planning

Distributed Systems: orchestration, phased rollout, sharding, observability, reliability, E2E integration testing, CI/CD

Ads Infrastructure: ad serving pipelines (publishing, matching, ranking), post-impression (attribution, conversion)

EDUCATION

University of Pennsylvania

Philadelphia, PA

Master of Science

Michigan State University

East Lansing, MI

Doctor of Philosophy

EXPERIENCE

Senior Software Engineer, Meta Platforms

2022 – Present

Sunnyvale, CA

IC5 · Greatly Exceeds Expectations (top 10% performer)

Unified E2E Test Platform — “TestPilot”

- Team lead of 10+ engineers from ads infra, ads products, and ads change management to deliver a unified E2E functional testing framework covering ads pre-delivery (publishing, indexing, retrieving, matching, and ranking) and post-delivery (impression, clicks, conversion, and revenue).
- Personally owned E2E TestPilot infrastructure architecture and drove key design tradeoffs across deterministic vs. non-deterministic delivery flow and joint vs. isolated test orchestration.
- Shipped TestPilot AI skill suite for test authoring, validation, and enablement, driving ads targeting E2E test coverage from 0% to 95%.
- Launched TestPilot for pre-production ad lifecycle validation, directly addressing *\$300M/year* in revenue exposure from reliability risks and product development inefficiencies.

Less Personalized Ads (LPA)

- Reliability area lead for LPA: built testing infrastructure from zero, where no prior reliability coverage existed, architected the full coverage model (canary, CI/CD integration, privacy API contracts).
- Drove launch-readiness decisions (gating criteria, failure modes, blast radius controls, exception handling), unblocking LPA launch and preventing *\$10M* revenue loss.
- Shipped observability platform for LPA covering 100+ delivery-critical metrics across pre-commit, deployment, and production stages; defined SLOs, dashboards, and alert thresholds to reduce incident risk and shorten detection/triage loops.

Third-Party Data (3PD)

- Architected and launched asset-level privacy policy enforcement with end-to-end propagation across ads serving infra, unblocking Meta’s 3PD privacy commitment at scale.

Software Engineer Intern, Oracle Corporation

05/2021 – 08/2021

Seattle, WA

- Built an event-driven E2E test automation pipeline for Oracle Hardware validation, triggering fleet-level validation on hardware events and eliminating manual test cycles for release qualification.